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Pursuing Upside Potential of Hybrid Low Salinity Waterflood EOR in Carbonate Reservoirs

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Abstract

In our previous LSW EOR research focused on fluid-fluid interaction (FFI) forming micro-dispersion (MD) as an interfacial reaction between oil and LSW (sea water, SW, diluted to 1%), we identified diethyl ketone (DEK) as an effective additive that can accelerate the interfacial reaction and improve oil recovery. The tertiary mode core flood of hybrid LSW EOR (containing 2wt% DEK) achieved +11% IOIP and +8% IOIP compared with secondary sea water injection and pure LSW injection, respectively. This paper pursued further upside potential of hybrid LSW EOR by exploring more effective additives to assist FFI. The exploration was carried out based on a simple evaluation of micro-dispersion ratio (MDR) that was defined as the increase in water-in-oil content resulting from oil-water interfacial reactions. The MDR, measured by FFI tests statically mixing oils and LSW, represented a key performance indicator highly correlated with oil recovery improvement because our previous work revealed a linear correlation between MDR and oil recovery increment. For this study, a total number of ten stock tank oils (STOs) were collected from multiple locations of sub-reservoirs in a target offshore carbonate field via two sampling campaigns. The first campaign collected four samples of STO-L1/L2/L3/U from mid-dip sections of lower (L) and upper (U) reservoirs in the field while the second campaign collected six samples of STO-L5/L6/L7/L8/L9/L10 from mainly down-dip sections of the L reservoir. The multiple locational sampling was deployed based on our previous findings in which a molecular-level fluid analysis using Fourier Transform Ion Cyclotron Resonance Mass Spectrometry (FT-ICR MS) detected involvement of heavy polar components into MD formation in the FFI mechanism. Therefore, deeper location oils containing rich heavy-ends were expected to demonstrate higher MDR because of compositional gradients in large structural depth variation in the target field.

In each campaign, the best positive oils showing the highest MDR (i.e., 20 v/v by STO-L2 and 199 v/v by STO-L5) at pure LSW were screened as reference for further hybrid LSW EOR potential survey. This clearly demonstrated that MDR can be higher in down-dip oils. Additive candidates were $CuCl_2$ and ketones such as dimethyl ketone (DMK), diethyl ketone (DEK), and methyl ethyl ketone (MEK). In addition, dimethyl sulfoxide (DMSO) was selected from the perspective of polar substances due to having larger dipole moments than the ketone candidates. Furthermore, a sensitivity analysis of additive concentration was conducted for techno-economic optimization. The reference MDR of mid-dip STO-L2 increased to 53 v/v at 2wt% DEK while the reference of down-dip STO-L5 increased to 300 v/v at 0.5 wt% DEK, 273 v/v at 1.0 wt% DMSO, and 330 v/v at 0.1 wt% MEK. The exploration with concentration sensitivity showed optimum concentration ranges that improve techno-economic potential (*e.g.*, six times MDR with one quarter DEK concentration). Our new efforts of more-affordable additive designs demonstrated increasing value of hybrid LSW EOR option.

Introduction

Yarlagadda et al. (2024) reported an outlook for future long-term liquefied natural gas (LNG) market growth. The outlook predicted growth in global gas trade through LNG and pipelines. LNG was considered to make up a dominant share of gas trade; therefore, new LNG export infrastructures were estimated to increase from 230 to 840 million tons per annum (MTPA) by 2050 across scenarios. More recent outlook by Shell (2025) estimated that global demand for LNG will rise by around 60% by 2040 because of economic growth in Asia, artificial intelligence (AI) impact, and efforts to minimize emissions in heavy industries and transportation. Similar projections of increased demand were reported by many other sources (Casagrande and Dallago, 2025; Connelly, 2024; J.P. Morgan, 2025; Yanagisawa, 2024). With projected higher demand for LNG, greater priority has been given to the sales of produced hydrocarbon (HC) gas. This driver works to reduce motivation to use HC gas for enhanced oil recovery (EOR) purposes. Even though HC gas injection has been most technically established method among various EOR options (due in large part to expected higher recovery factor improvement), water-based EOR is being given more attention as an alternative option. Compared with water-based EOR using chemicals such as polymer and surfactant injection methods, low salinity water (LSW) EOR has been investigated as a low-cost option that can minimize additional CAPEX in existing water injection facilities and keep OPEX low.

The LSW EOR research has been conducted in not only sandstone reservoirs (McGuire et al. 2005; Lagar et al. 2008; Lebedeva et al. 2009; Pu et al. 2010) but also carbonate reservoirs (Yousef et al. 2010; Austad et al. 2012; Zahid et al. 2012; Mahani et al. 2015, 2017). Through a long history in LSW EOR research, various types of recovery-driving mechanisms have been discussed such as ion exchange, wettability alteration, and/or rock dissolution. However, the research community has not reached a consensus on the mechanism. We set our LSW EOR target to an offshore giant carbonate reservoir in the Middle East and we have specifically focused on fluid-fluid interaction (FFI) between LSW and oil. The attention on FFI was motivated from the past research findings in other groups and ours which succeeded proving a good agreement between high FFI causing oils and noticeable oil recovery increment in core floods (Masalmeh et al. 2019; Alhammadi et al. 2021; Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024c). As a part of the key theory in FFI mechanism, micro-dispersion (MD) has been proposed by Sohrabi et al (2017), AlHammadi et al (2018), and Mehraban et al. (2020). The theory assumes that LSW droplets form a MD phase at the interface of water and oil while the droplets are surrounded by surface-active materials present in LSW-reactive crude oil (Mahzari and Sohrabi. 2014; Facanha et al. 2017; Mahzari et al. 2018). LSW-reactive oil shows high MD ratio (MDR) which represents water content increment (the ratio between the water content in the MD phase and that present natively in the crude oil) by the oil-water interfacial reactions while non-reactive one shows low MDR. Each of them is defined as positive and negative oils, respectively.

Through a screening study based on the FFI mechanism, we selected the best positive oil (STO-L2) due to the highest MDR 20.3 v/v among crudes collected from our target carbonate reservoirs (Hiraiwa et al. 2023). To improve MDR, we investigated Hybrid LSW EOR using several types of small additives and finally found the best case using diethyl ketone (DEK) which achieved more than 2-times increment to the MDR

52.6 v/v (Yonebayashi et al. 2024a, 2024b). The attractive MDR was achieved using a mixture of LSW (i.e., TDS 430 ppm approximately: 100 times-diluted sea water) and 2 wt% DEK. The corresponding tertiary mode oil recovery increments were confirmed to increase from +3.0 % in the pure LSW case to +10.7% in the hybrid LSW case through core flood tests. For further improvement of MDR, we have explored more effective additive candidates and optimized the current best design using the additive of DEK. The mainstream history of LSW EOR design modification consists of brine optimization and/or combination of other EOR methods. The former spikes certain ion-concentrations such as divalent ions (i.e., Ca^{2+}, Mg^{2+}, and SO_4^{2-}), in particular (Zhang et al. 2006; Fathi et al. 2012; Awolayo et al. 2014; Al-Hashim et al. 2015; Alshakhs and Kobscek. 2015; Alshaikh and Mahadevan. 2016). The latter assumes a combination with chemical EOR such as polymer flood and/or surfactant flood. Low-salinity surfactant (LSS) flood has been reported positively (Belhaj et al. 2022, 2023, 2024; Kalam et al. 2023); however, low-salinity polymer (LSP) has not been simply concluded positive due to adsorption and inaccessibility (Souayeh et al. 2022). More recently, a further combined process was reported that sulphated LSP flood increased oil recovery compared with that of high-salinity polymer (HSP) injection case (Song et al. 2024).

In this paper, we describe more effective additive exploration by extending to types of solvent from DEK to similar but different ones (*i.e.*, dimethyl ketone, methyl-ethyl ketone, and dimethyl sulfoxide). Furthermore, more optimum design of hybrid LSW was investigated for improving both technical and economic effectiveness. Both laboratory investigations were conducted through FFI tests from the perspective of MDR improvement.

Materials, Equipment, and Procedures

The study focuses on a Middle East offshore giant carbonate oil field which consists of two main oil producing reservoirs with finer definition of multiple sub-reservoirs. Oil production commenced in 1967. Pressure maintenance started in 1972 through dump-flood water injection followed by sea water injection and crestal gas injection. For sustaining oil production plateau and aiming to further ultimate recovery, various EOR options have been studied for each reservoir.

Fluid Properties (Oil, Brine, and Additive)

A series of four (4) stock tank oils (STOs) were collected in 2022 and another series of six (6) STOs were collected in 2024. All STOs were sampled from producers independently completed in each sub-reservoir. All the STOs are light oil with approximately 40°API gravity. The first sampling campaign (2022) was conducted by concentrating on sub-reservoir-dependency without finer depth/location-dependency ideas. The second sampling campaign (2024) was conducted by incorporating ideas of depth-dependency and areal location-dependency in certain sub-reservoirs. The sampling location is summarized in Table 1. Figure 1 illustrates well locations in each sampling campaign and heavy-end compositional gradients. Fluids from the down-dip sub-reservoirs had as much as 1.7 times the asphaltene content of those from the mid-dip sub-reservoirs.

Table 1—Summary of Stock Tank Oil Sampling.

Sample Year	Sample No.	Reservoir	Sub-reservoir	Area	Depth
2022	STO-U	Upper	2		Mid-dip
	STO-L1	Lower	U	North	Mid-dip
	STO-L2	Lower	M	North	Mid-dip
	STO-L3	Lower	L	North	Mid-dip
2024	STO-L5	Lower	U	West	Down-dip
	STO-L6	Lower	U	South	Down-dip
	STO-L7	Lower	U	North-East	Down-dip
	STO-L8	Lower	M	West	Down-dip
	STO-L9	Lower	M	East	Down-dip
	STO-L10	Lower	L	South-East	Mid-dip

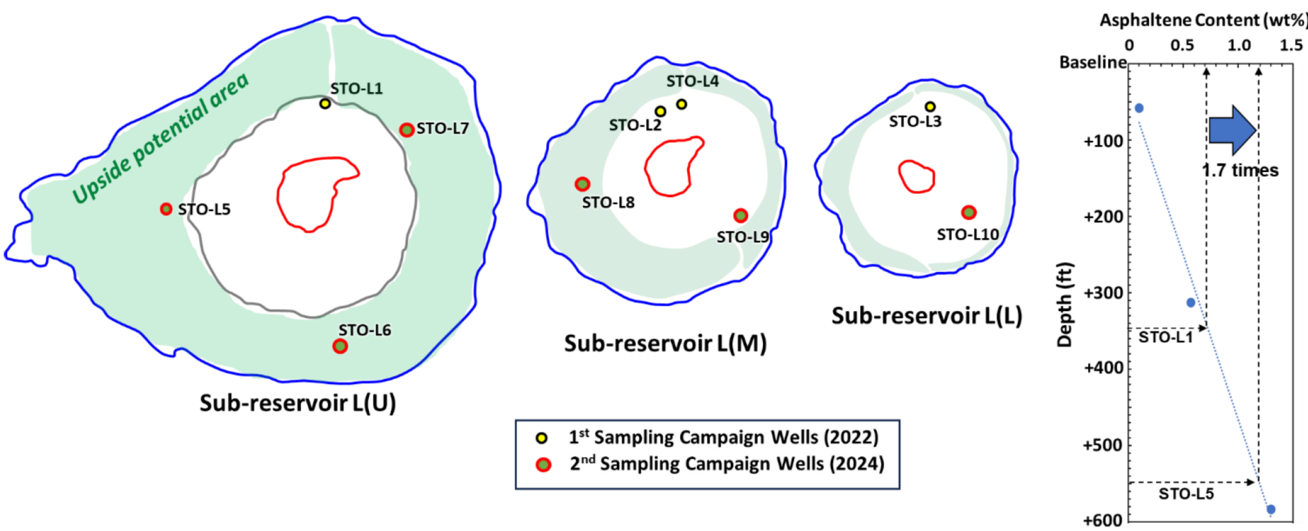


Figure 1—Sampling Location Map with Asphaltene Gradients in Lower reservoir.

The sampling idea based on depth/location dependency came from the previous study (Yonebayashi et al. 2024d). To identify surface-active components involved in FFI in the LSW EOR mechanism, the study applied petroleomics technology using Fourier transform ion cyclotron resonance mass spectroscopy (FT-ICR MS) (solariX 12T, Bruker Japan K.K.). This ultrahigh resolution technology is broadly used for hydrocarbon characterization and/or heavy oil research (Hsu et al. 1994; Rodgers et al. 2007; Marshall and Rogers, 2008; Hsu et al. 2011a, 2011b). For even further improved resolution, our FT-ICR MS analysis was combined with extended version of saturate, aromatic, resin, and asphaltene (SARA) analysis (Katano et al. 2020). Namely, we employed an independent application of FT-ICR MS analysis per each fractionated sample for seven fractions such as saturates (Sa), one-ring, two-rings, three-rings or higher rings aromatics (1A, 2A, and 3A+, respectively), polar (Po) and polyaromatics (PA) resins, and asphaltenes (As), *i.e.,* seven (7) analyses per sample. Consequently, we identified certain components in positive oils that assisted the FFI mechanism. The components were mainly polar heavy-ends such as asphaltene and resin; therefore, heavy-end-rich oil samples were considered from down-dip locations in the target field. Figure 2 summarizes the FFI mechanism elucidation study that became the backbone of sampling location strategy in this study.

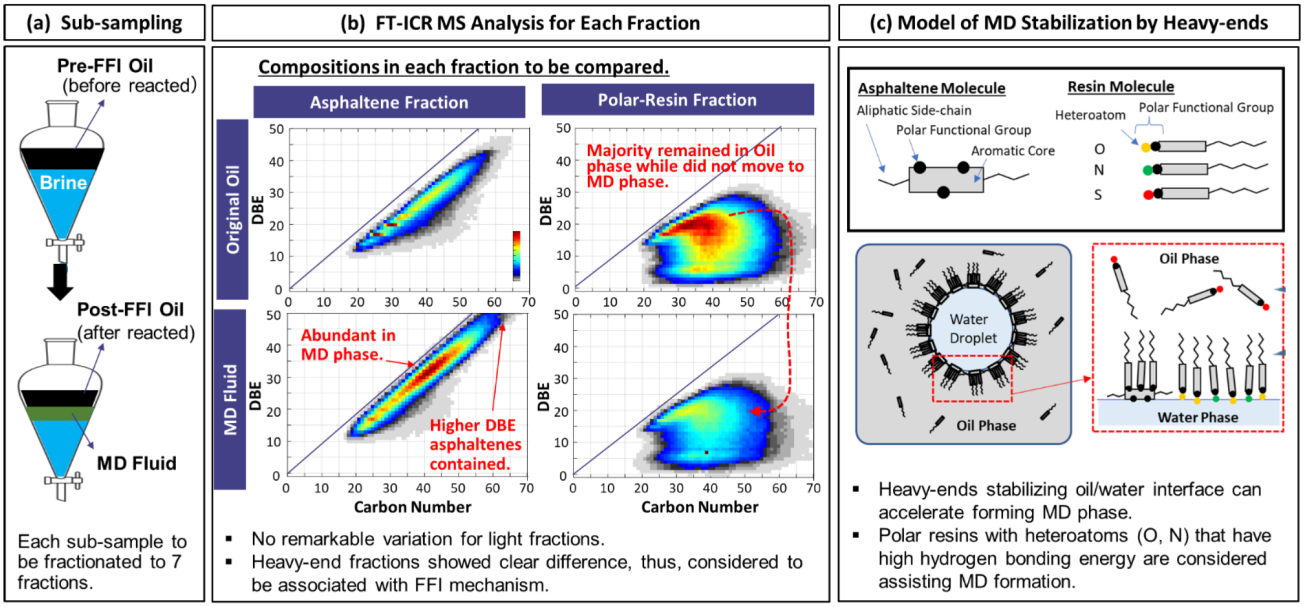


Figure 2—Identification of components involved in FFI mechanism: (a) Sub-sampling from original oil, MD fluid, and Post-FFI oil; (b) FT-ICR MS Analysis by Fractions with Heteroatom Classification; (c) Proposed model of MD stabilization by specified compositions.

Sample Collection and Fractionation (Figure 2(a)). A total of three samples such as original crude oil, oil/LSW interfacial fluid defined as MD fluid, and post-FFI oil were collected during FFI test. The collected samples were fractionated to a total of seven fractions to conduct FT-ICR MS analysis by each fraction for a more rigorous interpretation of double-bond-equivalent (DBE) vs. carbon-number (CN) plots. The residues in heptane were filtrated to separate soluble and insoluble components. The former components were maltenes (Sa, 1A, 2A, 3A, Po, PA) while the latter ones were dissolved in toluene and treated by Soxhlet extraction again to extract asphaltenes (As).

FT-ICR MS Analysis (Figure 2(b)). Each fraction was analyzed by FT-ICR MS method. Each analysis provided DBE-CN plot. Various combinations of the plots were compared by types of oil (i.e., positive oil vs. negative oil) and fractions. As a result, noticeable variations were found in the plots of heavy-ends (resins and asphaltenes) while not found in the plots of light-ends and intermediate components. Further detailed interpretation by heteroatoms (i.e., nitrogen, oxygen, and sulfur) showed that resin molecular transferring from original oil to MD fluid was abundant for nitrogen- and/or oxygen-containing components than sulfur-containing one. The essence of the findings was consistent to the previous studies for heteroatoms in heavy-ends involved in hydrogen bond formation with the H atom in the water molecule (Mizuhara et al. 2020; Ma et al. 2021; Fujita et al. 2022) revealing higher hydrogen-bonding energies in nitrogen/oxygen compared with sulfur.

Proposed Mechanism (Figure 2(c)). In our interpretation, components that showed no variation among samples were unrelated to the MD formation mechanism. On the contrary, we judged components observed abundantly in a specific sample were potentially involved to assist forming MD fluid. Under a general behavior varying surface charges of crude oil by salinity, the surface-active components diffused in the initial oil phase while in contact with high-salinity water (HSW) start gathering in the vicinity of oil/water interface when oil contacts LSW (Mehraban et al. 2021). The water droplet penetration into the oil phase triggers formation of reverse-micelle structures at the oil/water interface which is defined as MD (Emadi and Sohrabi, 2013; Sohrabi et al. 2017; Mahzari et al. 2019). The MD spontaneously forms by assistance of surface-active components such as asphaltene molecules containing polycyclic aromatic rings and naphthenic acids (Sanyal et al. 2017; Subramanian et al. 2017; AlHammadi et al. 2018) and N/O-containing polar resin components.

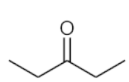
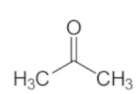
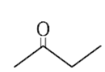
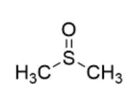
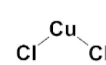
A total of three (3) brines were prepared as formation water, injection sea water (SW), and low salinity water (LSW). SW was reproduced the injection water which has been currently used in the water injection scheme (conditioned as specific gravity of 1.032 at 20 °C; pH 8.1). The LSW was prepared by 100 times dilution of SW. For analysis of effective LSW concentration, additional brines with other dilution ratios were prepared as 1.5%, 2%, 3%, 4%, 5%, and 20%. The properties of main brines are summarized in Table 2. Intermediate dilution brines were prepared proportionally by adjusting ion concentrations for each target.

Table 2—Summary of Brine Compositions.

Ions			Formation Water (FW)	Sea Water (SW)	Low Salinity Water (LSW) Diluted to 1% SW
CATIONS	Sodium (Na^+)	mg/L	67,416	13,790	138
	Calcium (Ca^{2+})	mg/L	19,127	484	5
	Magnesium (Mg^{2+})	mg/L	2,101	1,590	16
	Barium (Ba^{2+})	mg/L	8.3	<1.0	<1.0
	Potassium (K^+)	mg/L	2,359	470	5
	Strontium (Sr^{2+})	mg/L	1,204	8.6	0.09
ANIONS	Chloride (Cl^-)	mg/L	146,782	23,260	233
	Bicarbonate (HCO_3^-)	mg/L	746	154	2
	Sulphate (SO_4^{2-})	mg/L	208	3,230	32
	TDS	mg/L	239,951	42,987	430

A total of three (3) ketone family additives, one (1) polar aprotic solvent, and one (1) divalent metal ion were used in the study. In addition to DEK and dimethyl ketone (DMK) that had already been investigated in the previous hybrid-LSW EOR studies (Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024c), methyl-ethyl ketone (MEK) and dimethyl sulfoxide (DMSO) were newly evaluated. Table 3 summarizes additives.

Table 3—Summary of Additives.

Ions		Diethyl Ketone (DEK)	Dimethyl Ketone (DMK)	Methyl-Ethyl Ketone (MEK)	Dimethyl Sulfoxide (DMSO)	Copper(II) chloride (CuCl2)
Boiling Point	°C	101	56.5	79.6	189	993
Formula		$(CH_3CH_2)_2CO$	$(CH_3)_2CO$	C_4H_8O	$(CH_3)_2SO$	$CuCl_2$
Specific Gravity		0.815	0.784	0.805	1.10	3.386
Molecular Weight		86.13	58.08	72.11	78.13	134.45
Dipole Moment		2.80	2.88	2.78	3.96	NA
Structural Formula						
Remarks		Fisher Scientific, purity > 99%, CAS# 96-22-0	Sigma-Aldrich, purity > 99.5%, CAS# 67-64-1	Sigma-Aldrich, purity > 99%, CAS# 78-93-3	Sigma-Aldrich, purity > 99.9%, CAS# 67-68-5	

Fluid-Fluid Interaction Procedure

Figure 3 summarizes the overall workflow and experimental procedures. Based on the workflow (Figure 3(a)), first, The FFI test selected LSW-reactive oils defined as positive oils while screened out non-reactive oils defined as negative oils, using pure LSW without any additives. Second, further FFI tests were conducted using the most positive oil to investigate and effectiveness of LSW-plus-additive solutions to evaluate their MDR-boost performance. The positive oil formed micro-dispersion (MD) phase at an interface of crude oil and brine under static conditions while the negative oil retained a clear oil/water interface. This screening method was proposed as a practical criterion and widely applied by industry to display high LSW EOR performance (Masalmeh et al. 2019; Aljaberi and Sohrabi. 2019; Alhammadi et al. 2021; Façanha et al. 2022; Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024c). From the perspective of quickness and reliability, in particular, the method has attracted attention in industry among operators with suitable candidate assets because the subsequent core flood evaluation for FFI-screened positive oils may promise sure increase of oil recovery. As a pre-treatment, we filtered candidate oils through 5 µm pore size filters and centrifuged at 1400 rpm for 60 minutes to remove free water and solid contamination. The water content in the pre-treated STO is defined as bond water that is used to calculate the ratio of water content in MD phase to bond water content. This ratio is defined as micro-dispersion ratio (MDR). The details of FFI test procedure are summarized below (Figure 3(b)).

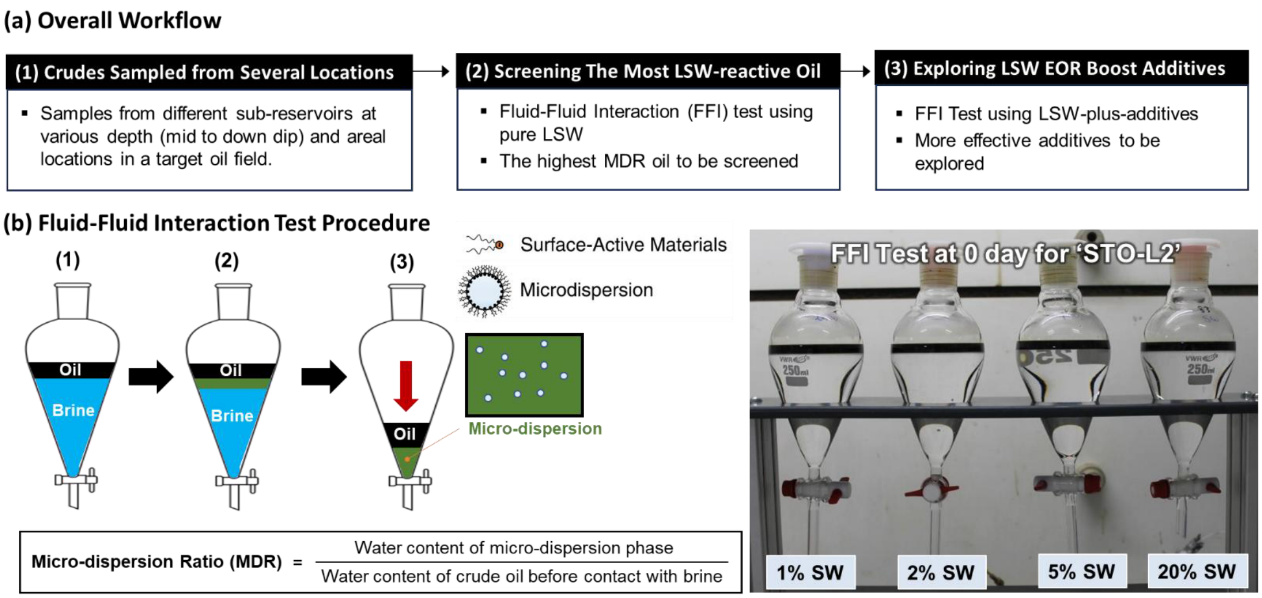


Figure 3—Study workflow and test procedures: (a) overall workflow; (b) fluid-fluid interaction test procedure and Photographs at 0 day.

1. Pour 230 mL brine into a separating funnel, then pour 30 mL crude oil into the separating funnel slowly with a pipette while avoiding any physical disturbance to the crude oil phase. Cap the funnel with a plastic quick-fit stopper to prevent evaporation.
2. Store the funnel in an oven at 140 °F (60 °C) for 3 days,
3. Remove aqueous phase slowly from the bottom of the funnel while avoiding physical disturbance of the oil-water interface. After cleaning outlet-line of the funnel with toluene and methanol, take sub-sample of crude oil (10-15 mL) from the interfacial area and store the sample in a 15 mL glass vial, then analyze water content of the sub-samples by Karl Fischer titration (Mettler Toledo Easy KFV Titrator).

Positive oil screening was conducted separately by sampling campaigns (2022 and 2024). In each campaign, the best positive oil was screened. Subsequently, a variety of additive sensitivities were investigated for the screened positive oils. An overall workflow is illustrated in Figure 4. To minimize a total number of FFI tests, a systematic narrowing down approach was deployed.

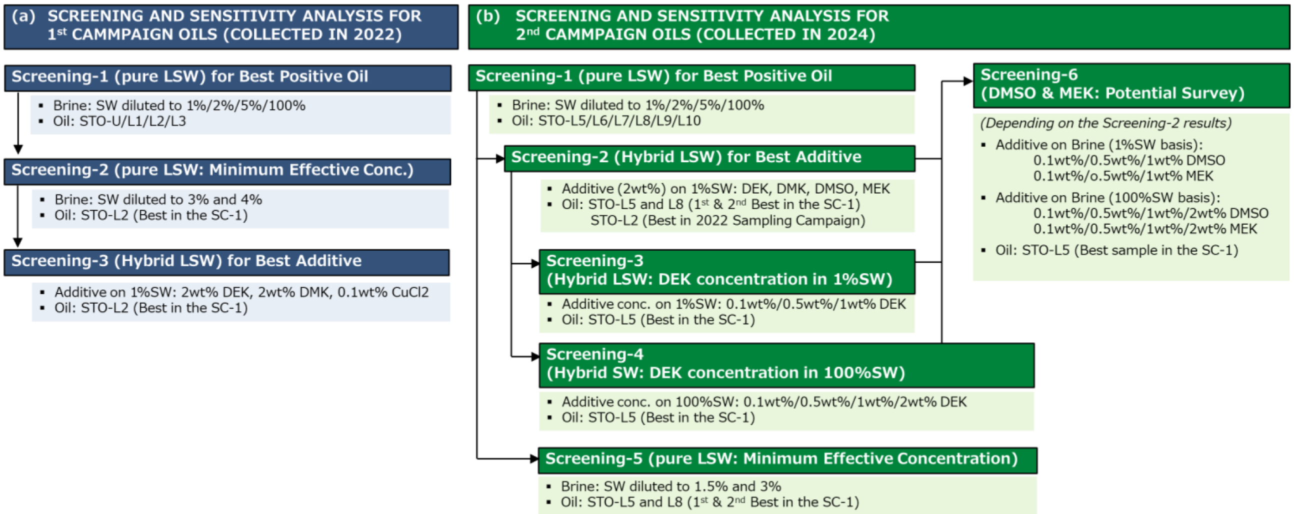


Figure 4—Overall Workflow of Screening and Sensitivity Study of Additives.

Fluid-Fluid Interaction

Screening Results of Positive Oil

Typical visual observation photographs are summarized in Figure 5 for STO-L2 contacting FW, LSW, and hybrid LSW. A clear separation was observed when STO-L2 contacted FW; however, a clear MD phase was seen at the oil/water interface when the oil contacted LSW (i.e., diluted to 1% SW). The color of MD phase seemed lighter than that of original oil. The surface of the interface between MD phase and LSW was fuzzy. The color and surface of MD phase became lighter and fuzzier in case of hybrid LSW.

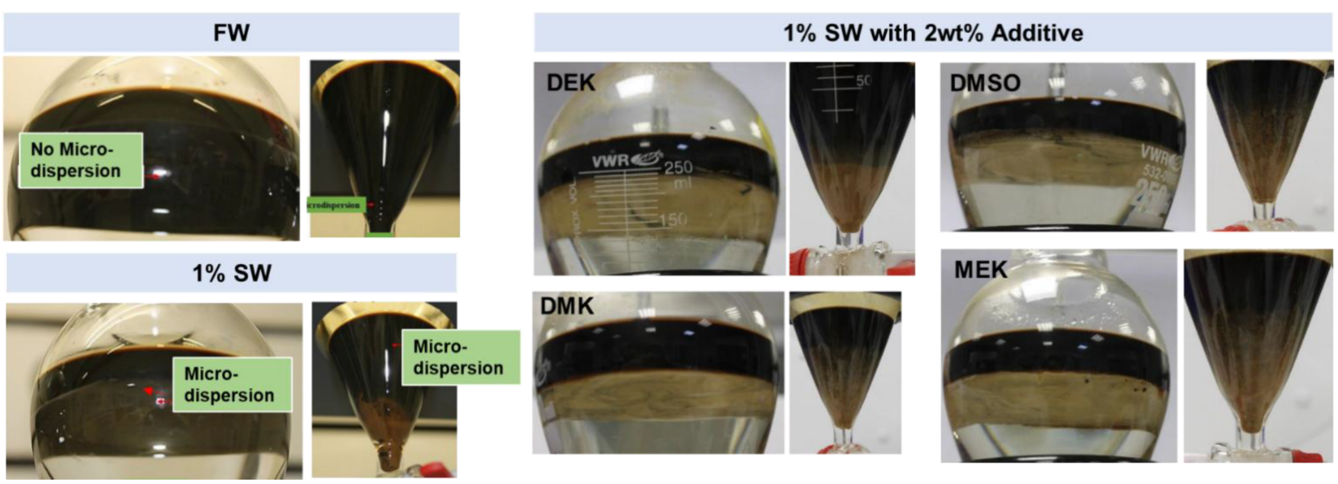


Figure 5—Visual Observation of FFI for STO-L2.

In a series of FFI tests for the first campaign oils, the lower reservoir originated oils of STO-L1, L2, and L3 were judged as positive oils because of exceeding a threshold value of MDR 5.0 v/v while the upper reservoir originated oil of STO-U was screened out as negative oil due to lower MDR than the threshold value (see Figure 6(a)). Among the three positive oils, the STO-L3 was considered lower potential compared with the other two because the effective LSW concentration was lower than that of STO-L1 and L2. The best positive oil was STO-L2 showing the highest MDR 20.3 v/v. Accordingly, the next sensitivity analysis of hybrid LSW was conducted using the STO-L2 at 100-times diluted concentration of SW (see Figure 6(b)). Among three additives of $CuCl_2$, DMK, and DEK, the highest MDR 52.6 v/v was obtained from hybrid LSW with 2wt% DEK. The MDR was more than twice that of the pure LSW (no additive) case. The additive sensitivity at both 100% SW and 1% SW showed clear positive impacts on MDR. While 2wt % DEK gave a larger relative improvement in 100% SW (maximum 3.7 times) than that observed in 1% SW (maximum 2.6 times), the highest *absolute* MDR was obtained for the 1% SW case with 2wt% DEK (52.6 v/v *vs.* 7.3 v/v for the 100% SW case). The boost of FFI by DEK was clear and more effective for the LSW system. Therefore, the most potent combination was concluded as hybrid LSW (1%SW and 2wt %DEK) for STO-L2 in the first campaign samples.

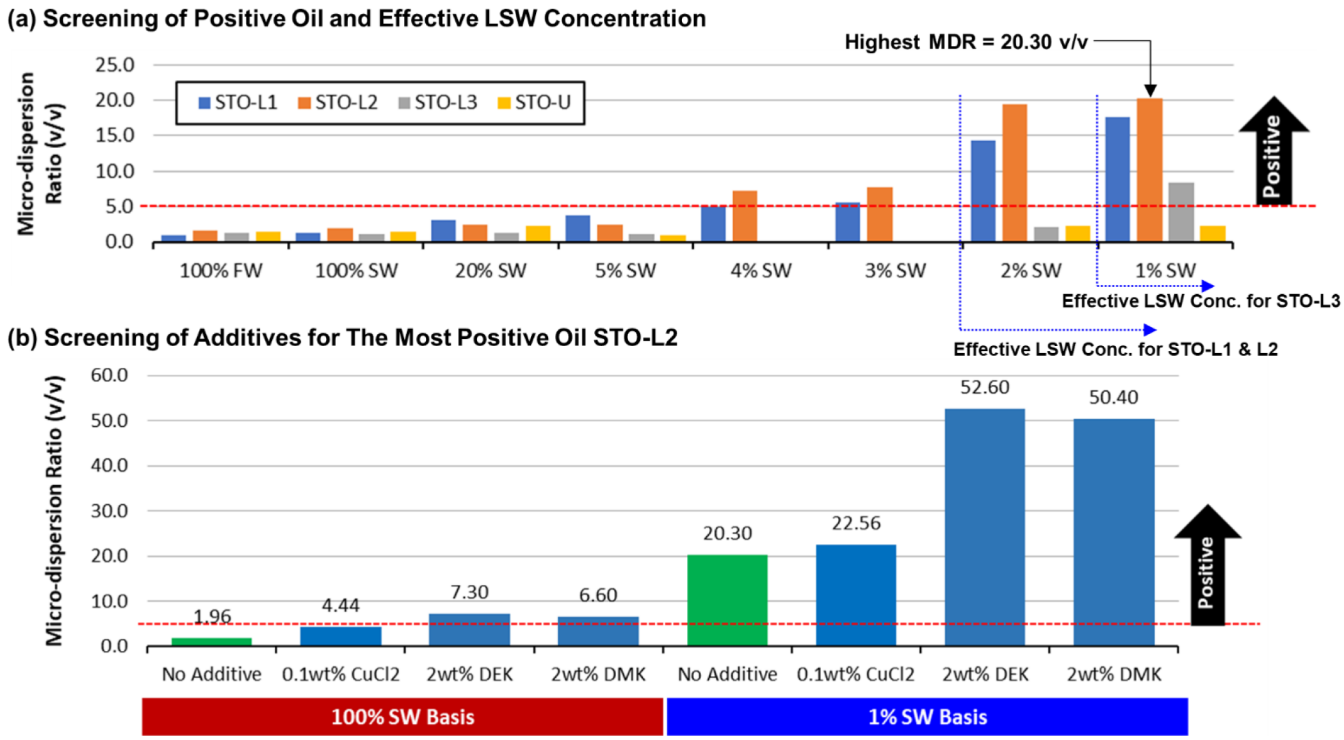


Figure 6—Fluid-Fluid Interaction Test Results for First Campaign Samples (2022): (a) STO-L2 screened as the most positive oil; (b) 2wt% DEK screened as the most effective additive to boost STO-L2 MDR.

A similar analysis workflow was applied for the second campaign oils in down-dip wells with extending types of additives as shown in Figure 7. The series of sensitivity analysis of LSW concentration (without additive) showed noticeably higher MDR compared with that of the first campaign investigation. All oil samples exceeded the threshold MDR (5.0 v/v) at 1%SW. The best positive oil of STO-L5 showed MDR 199 v/v that equals 10 times the best MDR 20.3 v/v of the first campaign screened STO-L2. Not only STO-L5 but also the other three oils exceeded the best MDR observed in the first campaign: *i.e.*, MDR 139.4 v/v by STO-L7, 172.1 v/v by STO-L8, and 74.7 v/v by STO-L9. Further sensitivities of additive and dose concentrations were investigated using the most promising oil, STO-L5. Regarding boost by additives, Figure 7(b) shows further enhancement from the baseline MDR 199 v/v of no-additive case in the LSW system. The best MDRs in each additive were 300 v/v at 0.5 wt% DEK, 273 v/v at 1.0 wt% DMSO, and 330 v/v at 0.1 wt% MEK. Figure 7(c) shows MDR enhancement from the baseline MDR 25.4 v/v of no-additive case in the SW system. The enhancement observation was a similar trend in the first campaign oils; however, the boost magnitude by additives was more significant in the LSW system compared with the SW system. Namely, the additives can work accelerating FFI more effectively in the LSW system for the down-dip STO-L5.

Figure 7—Fluid-Fluid Interaction Test Results for Second Campaign Samples (2024): (a) STO-L5 screened as the most positive oil; (b) Additive evaluation for STO-L5 at 1%SW; (c) additive evaluation for STO-L5 at 100%SW.

As shown in Figure 7(b), we judged that all three additives of DEK, DMSO, and MEK were useful to boost FFI between STO-L5 and 1%SW, in particular. A deep dive of interpretation was conducted for each additive to discuss optimum dose concentration as shown in Figure 8. Each graph extracts an optimum dose-concentration range for three additives. The ranges were around 0.5 wt%, 1.0 wt%, and 0.2 wt% for DEK, DMSO, and MEK, respectively. Instead of the single dose concentrations of ketone additives fixed at 2.0 wt% in the previous works (Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024c), the interpretation suggested a potential clue from the techno-economic perspective. In particular, we note that the single dose concentration fixed at 2.0 wt% was not best-performing even from a purely technical perspective, leading to the interesting conclusion that lower concentrations might be preferred for both technical and economic reasons.

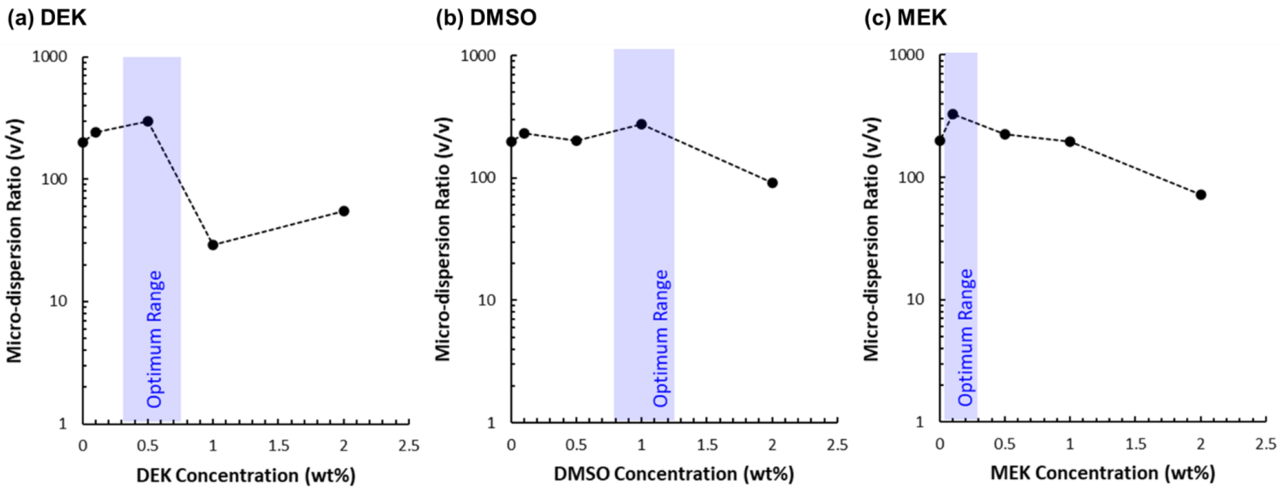


Figure 8—Optimum Additive Concentration Ranges for STO-L5 at 1%SW during Fluid-Fluid Interaction Tests: (a) DEK; (b) DMSO; (c) MEK.

A comprehensive comparison for overall additive sensitivity is shown in Figure 9. The baseline MDRs without additives became larger in the downdip oils (2024 campaign) than that in the mid-dip samples (2022 campaign). While MDR enhancement by adding 2wt% DEK was observed for STO-L2, but the same DEK concentration did not enhance MDRs from the baseline for the down-dip oils. As a consideration, there might be little room remaining for MDR enhancement due to higher baseline MDR in the down-dip oils, while there remains significant potential for enhancement in the mid-dip oil due to lower baseline MDR. As above-mentioned about dose concentration optimizing potentials, we are pursuing more optimum hybrid LSW design as a way forward.

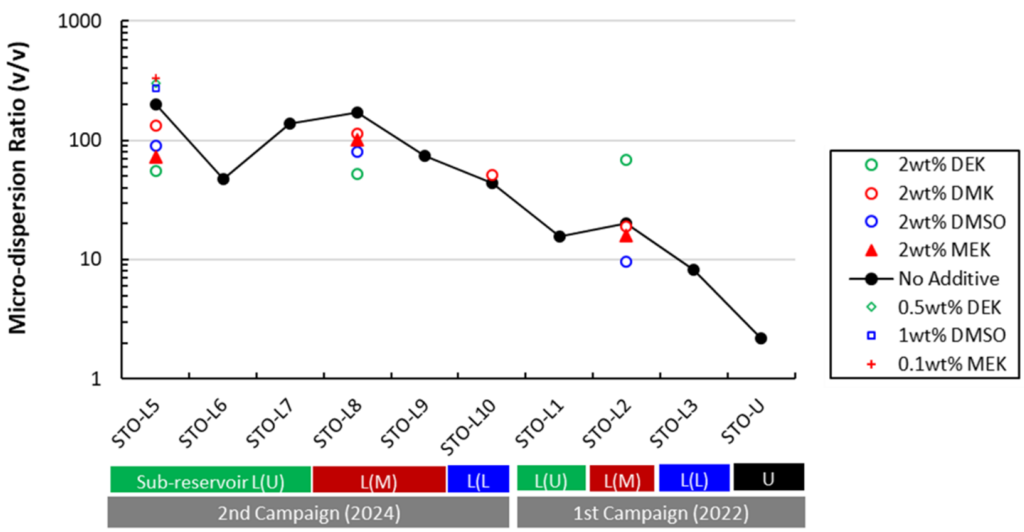


Figure 9—Overall Comparison of Additive Sensitivity at 1%SW during Fluid-Fluid Interaction Tests.

Way Forward of Core Floods

In the recent pure LSW EOR and/or hybrid LSW EOR studies targeting the same field (Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024c) using the first campaign oil samples, we found out a monotonic correlation between MDR and oil recovery increment by tertiary injection mode core floods as shown in Figure 10. A greater oil recovery increment is obtained from oils showing greater MDR. Our new study using the second campaign oil samples showed even higher MDR at pure LSW and hybrid LSW conditions. Therefore, a way forward will be core flood tests to evaluate qualitative oil recovery increment using the best MDR-showing oil at optimum dose concentrations.

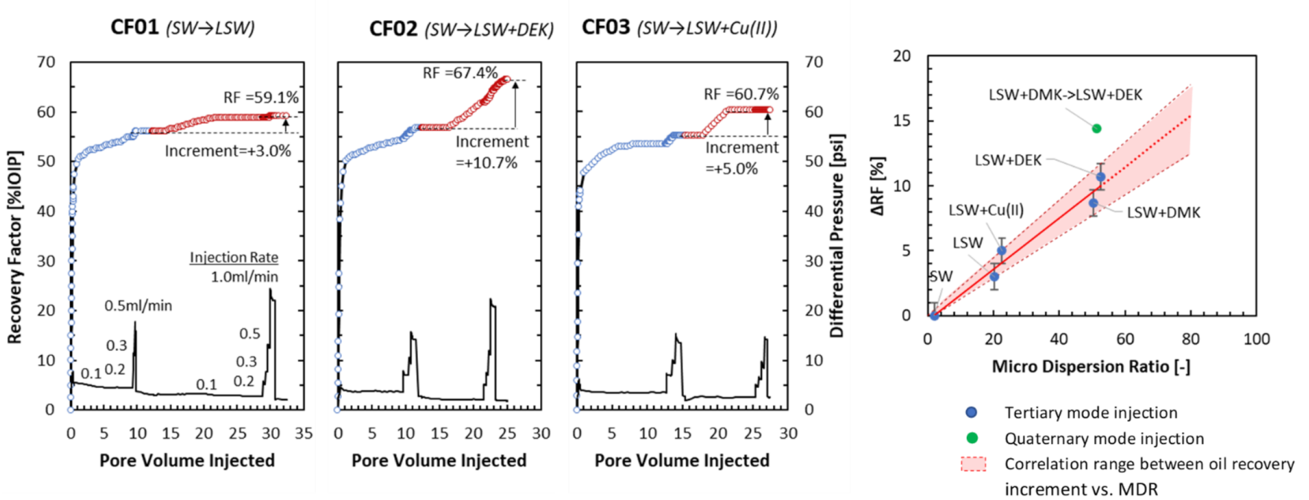


Figure 10—Core Flood Results and Monotonous Correlation between Oil Recovery Increment and MDR.

Conclusion

Based on the previous findings of a monotonic correlation between oil recovery increment and MDR defined as incremental ratio of water content in MD phase formed by interfacial reaction between the LSW-reactive positive oil and brine, we explored various additives into LSW to boost MDR through static FFI tests. The main findings were as follows:

- Two base positive oils of STO-L2 and STO-L5 were screened from the first and second campaign samples that were mainly collected from mid-dip and down-dip sections, based on the highest MDR of 20.3 v/v and 199 v/v at 1%SW (*i.e.*, reference MDR in the pure LSW system), respectively. This showed higher MDR in the down-dip oil than that in the mid-dip oil. This was consistent with the previous findings of polar heavy end components involved in MD formation. Namely, the more down-dip oil is, the more abundant polar heavy end components are.
- Based on the mid-dip STO-L2, hybrid LSW EOR potential was investigated for $CuCl_2$, DMK, and DEK. The FFI tests concluded the best additive as DEK at 2 wt% in the LSW system. The MDR increased more than two times to 52.6 v/v from the reference 20.3 v/v. On the contrary, based on the down-dip STO-L5, the hybrid LSW EOR potential analysis did not increase from the reference MDR at the same concentration of 2wt% DEK and other two additives. However, the subsequent sensitivity analysis suggested lower DEK concentrations had a MDR boost potential. The noticeable MDR 300 v/v was achieved at 0.5 wt% DEK. Similarly, DMSO and MEK boosted the MDR to 273 v/v and 330 v/v at 1.0 wt% and 0.1 wt%, respectively. This suggests the existence of optimum concentration and further techno-economic upside.
- The study highlights higher potential of hybrid LSW EOR for down-dip oil and by optimizing additive design. Three additives of DEK, DMSO, and MEK increased MDR to more than 1.3-1.7 times from the reference value through FFI tests at each optimum dose concentration in the down-dip oil and LSW system, implying a further incremental increase in the oil recovery factor could be achieved. A way forward of core flood test will be useful for future qualitative oil recovery increment evaluation using down-dip oils and optimum designed additives for further optimization of hybrid LSW EOR. Considering the high MDR results obtained in the study in conjunction with

the correlation between MDR and oil recovery increment, more attractive oil recovery increment is expected compared with the past records.

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